Hardware Advisory Primer: Mobile Manipulation for Luxury Reverse Logistics

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Hardware Advisory Primer: Mobile Manipulation for Luxury Reverse Logistics

Prepared for Anshu Arora, 21 October 2026. Subject: advising Aqilis on hardware architecture and component selection for a deployment at The RealReal.

This primer assumes you already know what a mobile manipulator is, what a VLA policy does, and how an RL training loop is structured, so none of that is explained again. What follows is the part that decides whether a deployment works: the physics of holding a handbag without marking it, the specification lines that actually constrain a build, the integration seams where multi-vendor systems fail, and the economics that make a cheap arm expensive. Every figure is labelled either as a manufacturer specification, a published measurement, or an estimate from a secondary source, because a surprising fraction of what circulates as robotics pricing comes from content farms that generate plausible numbers. Where I could not verify something, the primer says so rather than filling the gap. One framing note before the index: The RealReal is not an Amazon-style fulfilment problem wearing different clothes. Amazon's manipulation problem is millions of repeated SKUs where per-item learning amortises; The RealReal fulfilled roughly 3.59 million orders in 2025, and its consigned inventory consists of items that are each, in the inventory sense, unique and never recur (SEC, The RealReal 2025 Form 10-K). The filing reports orders rather than units handled, so treat orders as a lower bound on item-level throughput. That single structural difference drives most of the hardware conclusions in this document.

On reading time. The full document runs about 13,500 words, which is 45 to 55 minutes at technical reading speed. The fourteen topics with five required elements each could not be covered honestly in less without thinning them into bullet points. If you have 35 minutes, read in this order and stop when the time runs out: section 1, section 13, section 14, then Appendix A and Appendix C. That path carries the argument and the decisions. Sections 2 through 12 are the supporting engineering detail, and they are written to be read out of order, so treat them as reference during and after the meeting rather than as a sequence.

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1. Reverse-logistics task analysis

What is the actual work at The RealReal, stated as a manipulation problem?

The RealReal's own filing describes the flow: consigned goods arrive by at-home pickup, drop-off, or prepaid shipping into authentication centres, where gemologists, horologists, and brand specialists inspect items for markings, date codes, serial tags, and hologram stickers, supported by AI and microphotography for handbags and gemological instruments for jewellery; items are then written up, photographed, priced, stored, and later fulfilled (SEC, 2025 Form 10-K). The company reported 3,140 full-time equivalent employees, gross merchandise value of $2,130,007 thousand, 3,587 thousand orders, and an average order value of $594 for 2025, against a net loss of $41.8 million (same filing). Two sites carry the physical load, roughly 500,000 square feet in Perth Amboy, New Jersey and roughly 600,000 in Phoenix, Arizona, both of which a materials-handling integrator retrofitted with polybag ship sorters, conveyance, and put walls beginning November 2022 (VARGO press release).

Decomposed into manipulation primitives, the work is: unload and singulate inbound parcels; open or remove packaging; extract an item from a bag, box, or dust bag; present the item to a camera and to a human expert in a specified pose; place it onto a hanger, into a tote, or onto a shelf; retrieve it later by location; and bag, box, and induct it for outbound. The intake half is where the labour sits and where the variability is extreme.

Which of those are genuinely feasible on 2026 hardware?

Feasible today, with commercially supported hardware: parcel singulation and tote-level transport; picking rigid, boxed, or polybagged items from a bin into a tote; transporting garments already on hangers; placing items into fixtured positions; and presenting a barcode or label to a scanner. The reference point for the bin-picking case is Dex-Net 4.0, which on a two-gripper physical robot cleared bins of up to 25 novel objects at greater than 95% reliability and more than 300 mean picks per hour, using the suction cup on

82% of grasps (Mahler et al., Science Robotics 2019). That is a seven-year-old result and modern grasp models beat it, but it remains the cleanest published statement of what suction-first ambidextrous picking buys you on novel rigid objects.

Marginal, meaning demonstrable but not yet reliable enough to run unattended: grasping a single garment from a pile; handling handbags with chains, straps, and hardware that tangle; picking loose shoes, especially unpaired; and anything fragile with a protruding clasp or stone setting. Amazon's own positioning is instructive about the ceiling here. Amazon says Vulcan, its touch-sensing pick and stow robot, handles approximately three quarters of the item types in its fulfilment centres at speeds comparable to its operations employees, with the pick variant limited to items up to five pounds and 14 inches (Amazon, Introducing Vulcan; FreightWaves). Amazon has effectively unlimited capital, a captive data flywheel, and a far easier object distribution, and it reports three quarters.

Hard, meaning open research: removing a garment from a polybag or dust bag; laying a garment flat and wrinkle-free for photography; inserting a hanger into a garment; folding; untangling jewellery or chains; and any judgement of condition or authenticity by touch. The academic state of the art is blunt about this. A February 2026 systematic review of deep-learning cloth manipulation covering 41 papers from 2019 to 2024 found that most work addresses a single task using small towels or napkins, that only a minority address large cloths or complex garments, and that no universally recognised public dataset exists in the field (Frontiers in Robotics and AI, 2026). When the benchmark literature is still working on towels, a silk blouse in a crumpled polybag is not a near-term deployment target.

What is the hidden trade-off that almost everyone gets wrong here?

Throughput is the wrong headline metric for this customer. With an average order value of $594 and an inventory of unique, high-value, visually graded goods, the dominant cost of a manipulation failure is not the lost cycle, it is the marked, scuffed, or crushed item that drops a grade or becomes unsaleable. A luxury resale item's value is substantially its condition grade. A system that picks 20% faster but adds 0.2% damage on $594 average items is destroying value, and section 13 works that arithmetic explicitly. Engineers arriving from parcel or grocery automation import a throughput-first instinct that is actively wrong in this account.

Questions for Aqilis on this topic: Which specific station in the intake flow are they targeting, and did The RealReal choose it or did Aqilis? What is the agreed definition of a task failure, and does it distinguish a no-pick from a drop from a cosmetic mark? Have they been given a stratified sample of real items, or are they working from photographs and category lists?

2. Mobile manipulator architecture

What is the actual architectural decision, stated cleanly?

A mobile manipulator is a serial chain of error sources: base localisation, base-to-arm mount, arm kinematics, tool flange, end effector, and the item's pose within the grasp. Adding mobility inserts the single largest error term in that chain. A MiR250 docks to a VL-marker at plus or minus 3 mm in X and Y and plus or minus 0.5 degrees in yaw (Mobile Industrial Robots, MiR250 specifications). A UR10e repeats its pose to

plus or minus 0.05 mm per ISO 9283 (Universal Robots collective datasheet). Mount the second on the first and the system does not repeat to 0.05 mm, it repeats to roughly 3 mm plus the lever-arm effect of the yaw error. At a 700 mm working radius, 0.5 degrees of yaw is about 6 mm of lateral displacement at the tool. So a mobile manipulator is only viable if perception closes the loop at the point of grasp, which in practice means a wrist camera and a visual-servo or re-localisation step after every dock. If the architecture assumes the base positions the arm, it will not work.

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ERROR BUDGET, FIXED STATION vs MOBILE STATION

 fixed:   [ bolted frame ]---[ arm +/-0.05mm ]---[ EOAT ]---> ~1-2 mm at item
                                                  (hand-eye calib dominates)

 mobile:  [ dock +/-3mm, +/-0.5deg ]---[ arm ]---[ EOAT ]---> ~8-12 mm at item
             |                                                   (dock dominates)
             +--> 0.5 deg x 700 mm reach = ~6 mm lateral

 fix: wrist camera re-localises target AFTER dock, every cycle.
      cost: +0.2 to 0.8 s per cycle, +1 camera, +hand-eye calibration burden
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How do the four configurations actually compare?

A fixed-base arm is the cheapest, stiffest, fastest, and easiest to make safe, and it is the right answer whenever the work can be brought to the robot by conveyance. The RealReal already has conveyance and put walls at both sites, which means the work is already being brought to stations. A wheeled mobile manipulator buys access to many locations with one arm, at the cost of the error budget above, battery management, docking infrastructure, floor-space negotiation with human workers, and a second safety regime. A dual-arm system buys bimanual tasks that are genuinely impossible single-handed: holding a bag open while extracting an item, tensioning a garment between two grasps to flatten it, stabilising while unclasping. This matters enormously for the specific tasks listed as hard in section 1, because most of them are bimanual for humans too. A humanoid buys reach into human-designed spaces and a single embodiment across many tasks, and pays for it in stability, cost, serviceability, and immature safety certification. Agility's own public position has Digit at a 35-pound carrying capacity with a four-hour battery, with Digit 5 announced at 50 pounds and a 20-hour operating day and early access expected in the first half of 2027 (Agility Robotics); current-generation units work in segregated zones. For a facility whose entire value proposition is careful handling of fragile luxury goods near trained human experts, a legged platform in 2026 is a liability, not a feature.

Which specifications constrain the build, and how do you read them?

Payload on an arm datasheet means payload at the tool flange including the end effector, at full centre-of-gravity offset, across the whole workspace if the vendor is honest. Universal Robots states payload at full CoG offset and in the entire workspace, which is the strict reading and the one to insist on (UR collective datasheet). A 12.5 kg UR10e carrying a 2.5 kg gripper and a 1.5 kg camera has 8.5 kg of item budget, which is ample for apparel and handbags and irrelevant for the actual constraint, which is reach and cycle time. Reach is measured to the flange, so the effective reach is reach minus the tool length, and a long suction tool can add 200 mm of standoff that changes the workspace shape. Degrees of freedom above six

buy null-space motion for obstacle and self-collision avoidance in cluttered bins, which is why Boston Dynamics put seven axes on Stretch and says the extra axis helps avoid potential collisions and aids in placing cases in tight spaces (Stretch brochure). Footprint and stability for a mobile unit are a moment problem: payload mass times centre-of-mass height against wheelbase, evaluated under the worst-case deceleration, not statically.

What does the real-world example look like?

Boston Dynamics Stretch is the clearest commercially shipped instance of a purpose-built mobile manipulator for logistics, and its specification sheet is a lesson in scoping. It states a maximum case weight of 50 pounds (23 kg), maximum vertical and horizontal reach of 10.5 feet and 6.4 feet, a 40 by 48 inch footprint, 2,866 pounds of robot, 600 to 800 cases per hour, battery life up to 16 hours, 90% charge in 1 hour 45 minutes, and a narrowly defined eligible object class: rectangular cardboard boxes with taped or glued flaps, minimum surface length 6 inches, maximum 36 inches (Stretch brochure). A company with Boston Dynamics' engineering depth shipped a mobile manipulator by constraining the object class to one geometry with one surface type. That is the model to hold Aqilis to.

Questions for Aqilis: Is mobility required by the task or inherited from the platform they already have? What is the measured end-to-end pose error from dock to grasp, and what closes that loop? If the answer to the hard tasks in section 1 is bimanual, why is the architecture single-arm, or if it is dual-arm, what is the arm-to-arm calibration method and its residual?

3. End effectors and robotic hands

What is the honest ranking of end effector families for this inventory?

Suction is the workhorse and should be the default first channel. It is cheap, fast, tolerant of pose error, requires only one accessible flat-ish sealing surface rather than two opposing faces, and gets no credit for how much of the problem it quietly solves. Dex-Net 4.0's policy chose suction for 82% of grasps when given a free choice between suction and parallel jaw (Science Robotics 2019). For a facility whose inbound items arrive in polybags and whose outbound items leave in polybags, that matters doubly, because the bag itself is often a better suction target than the item inside it. Schmalz sells bag-specific tooling with a maximum suction area of 405 by 305 mm and load capacity up to 50 kg, with FDA-compliant silicone or wear-resistant NBR sealing surfaces (Control.com on Schmalz PSSG), and a ready-to-connect Bin-Picker SBPG for chaotic bin contents (Schmalz Bin-Picker SBPG).

Parallel-jaw grippers are the complement: they handle what suction cannot seal, which is porous fabric, mesh, textured leather, perforated material, and anything with no flat face. The two reference products are the Robotiq 2F-85, with 85 mm stroke and a programmable 20 to 235 N grip force (HowToRobot listing of Robotiq 2F-85), though note that secondary sources disagree on that force range, with at least one summary stating a 5 to 100 N user-selectable range instead, so confirm it against Robotiq's own datasheet before designing to it, and the OnRobot 2FG7, with roughly 11 kg form-fit and 7 kg force-fit payload, 38 mm total stroke, 20 to 140 N adjustable force, IP67, and about 1.1 kg of mass (OnRobot 2FG7). The IP67 rating on the 2FG7 is worth noting because dust from textiles is a real wear mechanism. OnRobot's 3FG15

three-finger centric gripper offers up to 15 kg form-fit payload over a 150 mm stroke with external gripping diameters of 4 to 152 mm (OnRobot 3FG15), which is the right shape for cylindrical and rotationally symmetric objects such as bottles, watch cases, and rolled goods.

Adaptive and underactuated hands sit between a gripper and a hand. They conform to shape with fewer actuators than joints, which gives shape tolerance cheaply, at the cost of controllability: you cannot command a specific contact configuration, only a closing motion and a force. For unknown-geometry items this is often a feature. Multi-finger dexterous hands buy in-hand reorientation, fingertip-precision grasps, and the ability to execute human-like strategies, and they cost one to two orders of magnitude more while adding tendon wear, calibration burden, and many new failure modes.

When is a tool changer the right answer instead of a hand?

When the task set is separable and the tool-change cost amortises. The decision rule is a comparison of three times: the per-cycle penalty of carrying a compromise tool, the per-change cost of swapping, and the per-cycle gain of the right tool. If items arrive sorted or sortable into runs by category, a changer wins decisively, because within a run you pay no change cost at all. If items arrive fully randomised one at a time, a changer loses and you want either a combination tool or two arms with different tools. The combination tool is the configuration RightHand Robotics ships: its gripper combines suction and mechanical fingers, and its public best-case rate is up to 1,200 items per hour, which the company's CEO framed explicitly as a best case with two totes near each other (The Robot Report on RightPick 3). Note what that sentence concedes: the headline throughput figure is a tote-geometry-dependent best case, not a sustained rate.

A five-finger hand is justified only when the task genuinely requires in-hand manipulation, meaning the item must be reoriented without being released. Opening a clasp, turning a bag to present a serial tag, and extracting a garment from a bag are plausibly in that class. Everything else at The RealReal is pick, move, place, and is better served by a cheap, reliable, interchangeable tool.

What is the common engineering mistake here?

Specifying grip force as a maximum rather than as a controllable range with a resolution, and ignoring the minimum. For fragile and markable goods the binding specification is the lowest force the gripper can reliably hold, and its repeatability at that setting, not the 235 N ceiling. A gripper that can only modulate force in 20 N steps is unusable on a soft leather bag no matter how adjustable its datasheet claims to be. The second mistake is forgetting that a vacuum system is a system: cups, fittings, tubing volume, generator, and valve response all set the achievable pick rate, and a 300 ms valve and a long tube will cap throughput regardless of how fast the arm moves.

Questions for Aqilis: What is their current end effector, and did they select it or inherit it? What minimum controllable grip force can they achieve and with what repeatability? Have they measured seal rate and hold time on real polybags and real leather, and if so what fraction of a stratified item sample was ungraspable by their tool?

4. Manipulation physics

What do I actually need to recall about grasp mechanics to ask good questions?

The useful working model is the friction cone and the grasp wrench space. At a contact with friction coefficient mu, the contact force must lie within a cone of half-angle arctan(mu) about the surface normal, or the contact slips. A grasp achieves force closure when the set of wrenches (forces and torques) the contacts can jointly exert spans the full six-dimensional wrench space, meaning any external disturbance can be resisted. Form closure is the stronger, friction-independent version achieved by pure geometry. Two important practical consequences follow. First, force closure is not binary in practice, it is a margin, and the useful question is how large a disturbance wrench the grasp resists before slipping, which is why acceleration limits and grasp quality are the same conversation. Second, soft contacts change the problem qualitatively: a compliant fingertip makes an area contact rather than a point contact, which adds a torsional friction term and dramatically improves resistance to rotation about the contact normal. That is most of why soft and underactuated fingers outperform their contact count.

How does suction physics differ, and what is the arithmetic?

Suction does not form closure at all. It generates a single normal force by pressure differential and resists tangential load only by friction against that normal force, and resists moments only by the cup's own stiffness and area. The lift force is the pressure differential times the effective sealing area: a 40 mm diameter cup has about 1.26e-3 square metres of area, so at a differential of 60 kPa the theoretical normal force is roughly 75 N. Real systems derate that heavily, commonly by half or more, for seal imperfection, dynamic loading, safety factor, and the fact that the item is rarely lifted purely normal to the cup. The governing failure mode is leakage, so the design variables that matter are porosity of the surface, surface roughness, cup material and lip geometry, and the flow capacity of the generator relative to the leak rate. This is why a porous knit garment defeats suction while the polybag around it does not, and why dust loading degrades suction performance over a shift in a textile environment.

What does the sensing side of contact physics actually cost?

Here is the single most useful specification comparison in this primer. A UR10e's built-in tool-flange force sensing has a range of 100.0 N in x-y-z with an accuracy of 5.5 N and a precision of 5.0 N (UR collective datasheet). That accuracy figure is the point: 5.5 N is roughly the weight of a 560 gram object. Built-in flange sensing on a mid-size cobot cannot resolve the contact forces that distinguish gentle from marking on soft leather or silk. If the deployment's value proposition is careful handling, a dedicated wrist force/torque sensor is not an upgrade, it is a requirement. The ATI Axia80 is the common choice in this class, with 16-bit system resolution, Ethernet or EtherCAT communication at up to 8 kHz frame rate, 12 to 30 V DC supply, 0 to 65 degrees C operating range, and an IP60 rating (ATI Industrial Automation, Axia80 kit).

Tactile sensing is a step beyond and is less mature than its demo reel suggests. GelSight's DIGIT is commercially available with an estimated lead time of 4 to 6 weeks after order and with replacement gel cartridges available (GelSight, DIGIT product page), and GelSight states plainly that the sensor is not designed for quantitative measurements. Vision-based tactile sensors in this family typically run at tens of frames per second, and a secondary engineering guide argues that slip detection needs at least 100 Hz,

which would exceed DIGIT's frame rate (Robotics Center tactile comparison); treat that threshold as a rule of thumb rather than a measured bound, but take the direction seriously. Slip is a fast, high-frequency phenomenon and a camera-based sensor sampling at 60 Hz is detecting slip after it has happened, which is useful for retry logic and not useful for arresting a drop.

Where does compliance belong in the architecture?

Somewhere, and the question is only where. The options are passive mechanical compliance at the tool (a remote centre compliance device, a sprung mount, a soft cup, a foam pad), series elasticity in the joints, active compliance via admittance or impedance control on a stiff arm, or compliance in the grasp itself via soft fingers. Each has a bandwidth and a cost. Passive compliance is free, instantaneous, and cannot be turned off. Active compliance is tunable and is limited by sensing bandwidth and loop rate, and crucially, by the force-sensing accuracy discussed above. The failure mode to design against is the stiff arm with coarse force sensing commanding a position into a rigid item, which is how expensive goods get crushed.

Questions for Aqilis: What force resolution do they have at the contact, and is it from the arm's flange or a dedicated sensor? Do they have slip detection, at what bandwidth, and does it drive a reflex or only a log entry? Where is the compliance in their stack, and what happens on a position command into an unexpected rigid contact?

5. Robot arms and actuation

What is the right mental model for the actuator choice?

First, a terminology correction worth making precisely because it recurs in vendor conversations: servo versus BLDC is a category error. A servo is a control architecture, a motor plus feedback plus a closed loop; brushless DC is a motor construction. Essentially every modern industrial or collaborative robot joint is a brushless permanent-magnet machine inside a servo loop, so the real question is never servo or BLDC, it is what reduction, what feedback, and what torque path.

The reduction choice is the decision that propagates everywhere. Strain-wave (harmonic) gearing gives high ratio in a compact, coaxial, light package with very low backlash, commonly cited below 0.1 to 1 arcminute when new, and its weakness is the flexspline under shock and its gradual loss of that backlash figure with wear. Cycloidal and RV gearing distributes load across many contact points and therefore tolerates shock far better, at typically higher new-condition backlash, cited in the range of 1 to 3 arcminutes for RV units. These figures come from vendor and engineering-blog comparisons rather than peer-reviewed test data and the sources disagree with each other, so treat the ordering as reliable and the specific numbers as indicative (IMSystems comparison; EVS Int industrial reducer comparison). Quasi-direct drive uses a low ratio, typically in the 6:1 to 10:1 range, with a large high-torque-density motor, trading absolute precision and holding torque for backdrivability and bandwidth, which is why it dominates legged robots and is rare in precision manipulators.

Which specifications matter for picking, carrying, and placing specifically?

Repeatability is the specification everyone quotes and it is the least relevant one here, because at plus or minus 0.05 mm every candidate cobot arm is far better than the pose uncertainty of the item being grasped. What matters instead: continuous versus peak joint torque and the duty cycle between them, because a pick-place duty cycle with high accelerations is thermally demanding; thermal derating behaviour, which Universal Robots states explicitly as 0 to 50 degrees C ambient with the note that above 35 degrees C the robot may run at reduced speed and performance (UR10e technical specifications); maximum joint speeds, which for a UR10e are 120 degrees per second on base and shoulder and 180 degrees per second elsewhere, and 120 to 210 degrees per second across the UR20's joints; and settling time after a move, which is the specification nobody publishes and which often dominates cycle time on a long-reach arm carrying a heavy tool.

Backdrivability deserves its own line. A high-ratio geared joint is not backdrivable, which means an unexpected contact is resisted rather than absorbed, and the item takes the energy. Series-elastic actuators insert a compliant element and measure its deflection to obtain joint torque, which gives genuine torque control and inherent shock tolerance, at the cost of control bandwidth and position precision. For handling fragile goods the ideal is high stiffness for positioning with fast, well-sensed compliance on contact, which in practice means a stiff arm plus a good wrist sensor plus a fast control loop, rather than a compliant arm.

What is the hidden trade-off?

Arm durability in this application is governed by the duty cycle you actually run, not the published MTBF. A pick-place workcell running 16 hours a day performs a few hundred thousand near-identical motions per month, and the wear concentrates in the two or three joints doing most of the work and in the end effector, which will fail long before the arm. Budget for end effector consumables as a line item: suction cups and tactile gel pads are wear parts. GelSight explicitly sells replacement gel cartridges for exactly this reason (GelSight DIGIT).

Questions for Aqilis: What joint-level torque sensing exists, if any, and is it from current sensing, series elasticity, or a dedicated sensor? What is the measured settling time at full reach with their tool mass? What is their expected duty cycle, and what in the arm or tool do they expect to replace first?

6. Mobile bases and navigation

Differential drive or omnidirectional, and what is the real trade?

Differential drive is simple, efficient, odometrically well-behaved, and cannot translate sideways, so it needs room to manoeuvre and it approaches a dock along a constrained path. Mecanum and omnidirectional bases translate in any direction, which makes dense aisle work and fine alignment to a station far easier, and they pay for that with mechanical complexity, higher rolling resistance, worse odometry because of roller slip, sensitivity to floor debris and seams, and typically lower payload-per-mass. In a facility with textile dust and polybag film on the floor, mecanum roller slip is a genuine reliability concern rather than a theoretical one.

Which specifications constrain the design?

Payload, speed, runtime, and docking accuracy, read together. The MiR250 is the useful reference because its public specification is unusually complete: 250 kg maximum payload, an 800 by 580 mm load surface, 300 mm height, 94 kg of robot, 2.0 m/s maximum speed, runtime up to 17 hours 30 minutes unloaded and up to 13 hours at maximum payload, a charge ratio up to 1:16 so that 10 minutes of charge yields 2 hours 40 minutes at maximum payload, docking accuracy to a VL-marker of plus or minus 3 mm in X and Y and plus or minus 0.5 degrees yaw, IP21, two SICK nanoScan3 safety laser scanners giving 360 degree protection, two 3D cameras, and eight proximity sensors (MiR250 specifications). Note the IP21 rating, which is indoor, dust-protected-against-fingers only, and note that the 250 kg payload figure is a transport rating for a flat load surface, not a rating for a 30 kg arm mounted on a mast generating dynamic moments. The MiR600 steps up to 600 kg on a 1,304 by 864 mm surface with an IP52 rating (MiR600 specifications), and the MC600 productised mobile manipulator pairs a MiR600 base with a UR20 or UR30 arm (Automated Warehouse on MC600).

How should I think about operating around human workers?

As a standards and liability question before an engineering one. The governing American framework is the three-part ANSI/A3 R15.08 series for industrial mobile robots: Part 1 covers the robot, Part 2 covers the integrated system and application, and Part 3, published in 2026, covers the use of IMR applications, including risk assessment, management of change, employee training, and periodic review (A3 on R15.08-3; ANSI Blog on R15.08-1). ISO 3691-4 is the internationally referenced analogue, and the MiR250 declares conformity to ISO 3691-4 with stated exceptions, ISO 13849-1, ISO 13850, ISO 12100, ITSDF B56-5, and RIA R15.08-1 (MiR250 specifications). The structural point that matters commercially: Part 2 places obligations on the integrator and Part 3 places obligations on the operating facility. If Aqilis deploys a mobile base into The RealReal, somebody is signing the application-level risk assessment, and that allocation should be explicit in the contract long before it is explicit in an incident report.

What is the common mistake?

Treating the published maximum speed as an operating speed. Speed near people is bounded by stopping distance, which is bounded by the safety scanner's protective field, which shrinks the usable field as speed rises. In practice a base around human workers runs well below its datasheet speed, which means cycle-time models built on datasheet speed are optimistic by a large factor. The second mistake is ignoring charge logistics: a 1:16 charge ratio is excellent, but it still requires dock availability, and a single dock serialises a fleet.

Questions for Aqilis: What speed do they actually run near people, and what protective field configuration produced that number? Who owns the R15.08 Part 2 and Part 3 obligations in the deployment? What is the tip-over margin with the arm extended at worst-case deceleration, and was it measured or calculated?

7. Sensors and perception hardware

What is the minimum viable sensor set per task?

For bin picking from clutter, the non-negotiable is a close-range 3D sensor over the bin with enough resolution to separate adjacent items, plus a wrist camera to re-localise at the grasp. For grasp execution on fragile goods, add wrist force/torque, and add tactile at the fingertips only if slip or contact-state estimation changes the control decision. For object verification, which at The RealReal means supporting authentication and condition grading, the critical hardware is high-resolution RGB under controlled, repeatable illumination, not depth. For placing, the requirement is usually the pose of the receptacle rather than the item, which a fixed camera or a fiducial solves more cheaply than a depth sensor.

How do I read depth camera specifications without being misled?

By working-distance band first and accuracy second. The Intel RealSense D405 has an ideal operating range of 7 cm to 50 cm with minimum object detection below 1 mm at 7 cm, which makes it a wrist camera and nothing else (Intel RealSense D405). It has no IR projector, so it degrades on low-texture surfaces, which in this application means plain-coloured fabric and smooth leather. The D435 family is quoted at under 2% depth error at 2 m, and the D455 at an ideal range of 0.6 to 6 m with under 2% at 4 m; both have minimum ranges in the tens of centimetres that disqualify them for wrist use, and the published minimum-distance figures disagree across Intel's own support posts and store listings, so verify against the current datasheet before committing a design. Structured-light industrial scanners sit in a different class: an integrator reports typical accuracy around plus or minus 0.1 mm with acquisition times of roughly 300 to 800 ms, which is the trade to internalise, since that scan time lands directly in the cycle (Relling, 3D vision cameras for bin picking). Zivid publishes unusually rigorous specifications that separate point precision, planarity, and dimensional trueness with stated test conditions, and recommends discarding the outer 5% of the frame because accuracy falls off at the edges (same source). Photoneo's MotionCam-3D targets moving scenes with parallel structured light and is now part of Zebra Technologies.

The cross-vendor comparison problem is worth stating explicitly because it will come up: optical 3D systems are specified against different standards, including VDI/VDE 2634, which measures probing error, sphere-spacing error, and flatness, so two vendors' accuracy figures frequently do not describe the same test (same source). Any spec-sheet comparison across vendors without reading the test conditions is noise.

What sensors are commonly bought and not needed?

3D LiDAR on a manipulator. LiDAR is navigation and safety hardware, and the safety-rated scanners on a base such as the SICK nanoScan3 are there for protective fields, not for grasp perception. Specifying an expensive spinning LiDAR for a manipulation cell is a recognisable sign that the sensor list was assembled by analogy to autonomous vehicles rather than from the task. The reverse mistake is skipping the IMU on a mobile base, where it materially improves odometry through slip, and skipping controlled lighting, which is the cheapest perception improvement available and routinely the highest-leverage one. Boston Dynamics ships onboard lighting on Stretch specifically so the vision system does not depend on trailer illumination (Stretch brochure).

Questions for Aqilis: What is in the sensor BOM, and for each item, which decision does it inform? What is their hand-eye calibration procedure, its residual error, and how often does it need redoing? How do they handle specular and transparent packaging, which defeats most depth sensors and is exactly what a

polybagged luxury item presents?

8. Hardware integration

Where do multi-vendor systems actually fail?

Not where people expect. The mechanical and electrical seams are largely solved by standards: the tool flange is specified by EN ISO 9409-1, with Universal Robots using the 50-4-M6 pattern on the UR10e and 80-6-M8 on the UR20, and tool I/O supplying 12 or 24 V at up to 2 A on a dual pin (UR collective datasheet). Those are a day of adapter-plate work. The failures are in three other places.

First, timing and control bandwidth. CAN is fine for a gripper or a sensor at tens to hundreds of hertz and is wrong for coordinated multi-axis control, because its arbitration and 8-byte frames do not give you deterministic synchronous updates across many nodes. EtherCAT with distributed clocks is the industrial answer for synchronised motion, and published results support sub-100-microsecond behaviour when the host is configured correctly: a dual-arm study using an optimised PREEMPT_RT kernel reported timing jitter under 60 microseconds with a 1,000 microsecond EtherCAT cycle and a 1,022.4 microsecond worst case (Chinese Journal of Mechanical Engineering, ROS2 real-time performance). The practical failure is in the software architecture rather than the bus: ros2_control users running at 1 kHz have reported multi-millisecond delays propagating from the control loop into the EtherCAT driver, with the suggested fix being a dedicated real-time thread for the EtherCAT exchange (ethercat_driver_ros2 issue 101). If Aqilis claims a sub-20-millisecond reflexive safety loop, the question is which thread, on which kernel, with what isolation, and measured how.

Second, calibration and coordinate conventions. Every vendor has its own frame conventions, units, rotation representations, and time base. Hand-eye calibration solving AX = XB is routine; keeping it valid across a tool change, a gripper swap, a base dock, and a thermal cycle is the actual work, and the residual error is what the grasp planner must tolerate.

Third, and most importantly, safety certification does not compose. Universal Robots certifies its arms to EN ISO 13849-1 PLd category 3 and EN ISO 10218-1 (UR collective datasheet), and MiR certifies its bases against ISO 3691-4 and R15.08-1. Wiring a PLd arm to a PLd base through a non-rated controller does not yield a PLd system. The integrated safety function has to be designed, validated, and documented as a system, and in a mixed-vendor build nobody is contractually obliged to do that unless somebody is named. This is where multi-vendor mobile manipulation projects quietly stall for months.

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INTEGRATION SEAMS, RANKED BY HOW MUCH THEY HURT

  mechanical flange      | ISO 9409-1 | adapter plate      | 1 day
  tool power / IO        | 24V, 2A    | wiring             | 1 day
  gripper comms          | CAN/RS485  | driver exists      | 1 week
  camera -> arm frame    | AX=XB      | recurring drift    | ongoing
  multi-axis sync        | EtherCAT   | RT kernel, threads | 1-2 months
  SAFETY as a SYSTEM     | 13849/10218| nobody owns it     | 2-6 months  <-- the killer
```

What is the hardware abstraction layer question?

Whether there is one, and who maintains the drivers. A hardware abstraction layer that covers arm, gripper, sensors, and base lets a policy be written once and retargeted, which is the whole premise of cross-embodiment learning. The cost is that the abstraction leaks exactly where it matters, at force, timing, and failure semantics. A gripper that reports “closed” at the abstraction layer may mean position reached on one vendor’s hardware and force threshold reached on another’s, and a policy trained against the first will fail silently against the second.

Questions for Aqilis: What is their hardware abstraction layer, and what does it abstract over today versus aspirationally? Who writes and maintains the drivers for components they do not build? Who owns the integrated safety validation for the deployed cell, and has a notified body or a safety consultancy been engaged?

9. Teleoperation and data collection

What does the data-collection hardware landscape actually cost?

The published anchors are: GELLO leader arms at roughly $300 per arm from the original 2023 publication, a ViperX-based bimanual ALOHA station at roughly $20,000, Mobile ALOHA at roughly $32,000, and UMI handheld grippers under $1,500 (DexSet rig comparison). Operating cost estimates from data vendors run $28 to $60 per raw teleoperation hour, with 10 to 30% of raw episodes failing quality review, and one source claiming 4 to 6 hours of human review per hour collected. Throughput estimates diverge wildly, from 2 to 4 demonstrations per hour for bimanual teleoperation to 30 or more per hour on tabletop tasks, which tells you the figure is entirely task-dependent and that anyone quoting a single number is selling something. These are commercial sources with an interest in the answer, so treat the hardware costs as solid, the per-hour costs as directional, and the throughput figures as unusable without your own pilot measurement.

Aqilis’s own white paper prices this the same way, anchoring teleoperation at roughly $28 per hour fully loaded, 20 to 40 demonstrations per hour for experienced operators on leader-follower rigs, and 20 to 300 demonstrations per skill, citing ALOHA and Mobile ALOHA reaching bimanual competence from 20 to 50 demonstrations and diffusion policies typically using 200 to 300.

What is the hardware-relevant part of the embodiment gap?

That a demonstration is recorded in one body and replayed in another. A human hand has roughly 20 or more degrees of freedom; a parallel-jaw gripper has one. Retargeting from hand tracking to a gripper is not a mapping problem, it is a lossy projection: the human strategy of rolling an object between fingers has no image in the gripper’s action space. This has a direct hardware consequence that is easy to miss. If you intend to collect human demonstrations and deploy on grippers, you must collect in a representation that the gripper can express, which is what UMI-style handheld grippers do by putting the target morphology in the human’s hand. If instead you collect rich human-hand data, you have committed to a multi-finger hand in the deployment, with everything that implies for cost and reliability. The data strategy and the end effector choice are the same decision, made once, and reversing it later invalidates the dataset.

Latency is the other hardware constraint. Contact-rich bilateral teleoperation degrades badly as round-trip latency rises, and the common engineering rule is that beyond roughly 100 to 150 milliseconds the operator switches from continuous control to move-and-wait, which changes both the demonstration's character and its usefulness as training data. If the rig is operated over a corporate network or from another site, measure the actual round trip under load before designing the collection programme.

What does Aqilis's architecture claim about this?

Something strong, and worth probing carefully. Their white paper's cross-modal learning section claims third-person monocular video of a human hand can be tokenised directly into the robot's internal coordinate space through a contrastive projection layer and internalised during an offline consolidation cycle, which they position as removing the need for teleoperation. If that works, the teleoperation rig drops out of the bill of materials entirely, which is a very large hardware and opex decision. The paper contains no empirical results supporting it. That is not a reason to dismiss it, but it is a reason to make the hardware plan robust to it being false, which means not building a deployment whose data strategy has no fallback.

Questions for Aqilis: What data have they actually collected, on what hardware, and how many hours of it? If cross-modal video learning works, why does the deployment need a robot-specific data path at all, and if it does not yet work, what is the fallback collection rig and its cost? What is the measured teleoperation round-trip latency on the intended network path?

10. Simulation and sim-to-real

What does simulation reliably buy, and what does it not?

It reliably buys reach and collision analysis, workcell layout, cycle-time estimation, kinematic feasibility, and training for anything dominated by geometry and gross dynamics. It does not reliably buy any prediction about whether a specific gripper will hold a specific handbag. The reason is structural rather than incidental: rigid-body engines approximate contact deliberately because solving it accurately is too slow to train against, and the approximation is weakest exactly at gripper contact.

The evidence is specific. MuJoCo models contact with a penalty-style soft constraint whose stiffness and damping are tuned per geometry, and a 2025 paper states the trade directly, that realistic hard contacts require very stiff settings which corrupt gradients, while softer settings strongly increase the sim-to-real gap (Hard Contacts with Soft Gradients, arXiv 2506.14186). A 2026 paper testing MuJoCo on block collisions found it captures linear velocity decay reasonably well but fails to reproduce angular velocity decay accurately, attributing the residual orientation error to inaccurate torsional friction effects (CLASH, arXiv 2602.18707). Torsional friction is precisely the term that governs whether a grasped object rotates in the gripper, which is the failure mode that drops items. A 2021 validation study on Cassie also found contact-parameter sensitivity varies by scenario, with friction mattering for sliding-heavy cube tosses and barely at all for mostly static jumps (Validating Robotics Simulators on Real-World Impacts, arXiv 2110.00541).

So what is the defensible workflow?

System identification against a small number of real trials, then randomisation around the identified parameters, then verification on hardware, with the honest expectation that transfer quality is task-dependent. Recent work identifies MuJoCo contact parameters (solimp, solref, friction) from as few as three real cube trajectories using CMA-ES, and the CLASH work identifies contact parameters from roughly ten real collisions and fine-tunes a learned correction on top of MuJoCo, reporting a doubling of success rate in sequential pushing with reinforcement learning (CLASH). These are preprints, and their generalisation to new hardware is not settled. A dual-simulator practice of training in Isaac Lab and re-scoring checkpoints in MuJoCo MJX before hardware rollout is emerging but is not an established standard, and MJX is not feature-identical with MuJoCo C, with gaps around tendons, composite objects, and mesh-mesh collisions.

What about deformables, which is the case that matters most here?

This is the weakest area of all. Garments require either finite-element or position-based-dynamics approaches that are slower and far less validated than rigid-body contact, and the cloth manipulation review noted earlier found no universally recognised public dataset and most work confined to towels and napkins (Frontiers 2026). A 2024 review adds that most sim-to-real efforts for textiles use domain randomisation without adaptation based on real-time feedback. The practical implication for an advisory conversation: if a vendor shows you a simulated garment-handling policy and projects a real-world success rate from it, that projection has no established basis. Ask for hardware numbers on real garments or treat the claim as unsupported.

Questions for Aqilis: Which simulator, and what system identification have they done against their actual hardware? What is their measured sim-to-real success-rate delta on a task they have taken all the way to hardware? For deformables specifically, is simulation in the loop at all, or is that entirely real-world learning?

11. Reliability, safety, and deployment

What is the current standards picture, stated accurately?

The collaborative-robot requirements that lived in ISO/TS 15066:2016 have been absorbed into the third editions ISO 10218-1:2025 and ISO 10218-2:2025, with the body-region force and pressure model carried forward; ISO's catalogue still lists the technical specification as current and marked for revision (InMotion on ISO 10218 and R15.06-2025; EVS Int, validating power and force limits). These are vendor and consultancy summaries rather than the standard text, so the structural claim is well supported and the detailed tables should be read in the standard itself. Two distinctions survive and matter operationally. Quasi-static contact, where a body part is trapped or clamped against a fixed surface, has lower permissible limits than transient contact, where contact is a brief impact, and a tool that passes a transient impact test can still fail a clamping scenario. And power-and-force limiting is a validated claim, not a design intent: it has to be measured with an instrumented device against the body-region limits for the specific tool, payload, speed, and geometry.

For mobile systems, the governing framework is the R15.08 three-part series described in section 6, with facility-level obligations in Part 3.

How should I think about uptime and failure recovery?

Availability is MTBF divided by the sum of MTBF and MTTR, and in deployed robotics MTTR is almost always the term that is mismanaged. A cell with excellent reliability and a four-hour response time for a part that is not on site has worse availability than a less reliable cell with a spares kit and a trained on-site technician. So the deployment questions are spares inventory, who is permitted to touch the cell, and what the documented recovery procedure is, not just the failure rate.

Failed-grasp recovery deserves to be treated as a taxonomy rather than one case, because the recovery differs for each. No grasp found means the perception or planning stage returned nothing, and the recovery is a bin shake, a different viewpoint, or escalation to a human. Grasp failed at closure is detected by gripper position or force and is cheaply retried. Item dropped in transit is the expensive case: it requires detecting the loss, localising the item, and in a luxury context, flagging it for condition re-inspection. Item placed incorrectly is a data-integrity problem as much as a physical one, because an item in the wrong location in a consignment business is an item that cannot be found for an order. Item damaged in handling is the case with no automated recovery at all, and the only real mitigations are preventive: force limits, compliance, speed limits, and conservative grasp selection.

What is the common mistake?

Measuring success rate without defining the denominator, and reporting pick success while excluding items the system declined to attempt. A system that attempts only the easy 60% of items and succeeds on 98% of those has a 59% effective success rate against the actual inbound mix, and both numbers are honest. In an advisory role, the single most useful question is always what fraction of presented items the system attempts, reported alongside the success rate on attempts.

Questions for Aqilis: What is their attempt rate and their success rate on attempts, measured on real inventory? What is their detection coverage for a dropped item, and what happens next? Has power-and-force limiting been validated with an instrumented measurement for their actual tool and payload, or is it a design claim?

12. Hardware vendor landscape

How should I read the pricing in this market?

Sceptically, and with a clear line between verified specification and street estimate. Almost no robotics vendor publishes list prices. During the research for this primer, a large share of search results for pricing came from aggregator sites that appear to generate plausible figures programmatically, and several of them contradicted each other and occasionally themselves within one page. Treat every dollar figure below that is not labelled as from a manufacturer as a street estimate to be confirmed by quote, and expect integration, tooling, and commissioning to add substantially. One price guide notes that accessories, integration, and training can double the base price of a cobot, which matches field experience (Standard Bots price guide).

Robot arms. Universal Robots is the default for this class of work because the specifications are published completely and the ecosystem is deep: UR10e at 12.5 kg payload, 1,300 mm reach, plus or minus 0.05 mm repeatability, 33.5 kg, 190 mm footprint, IP54; UR20 at 20 kg, 1,750 mm, plus or minus 0.05 mm, 64 kg, 245 mm footprint; all six-axis, all certified EN ISO 13849-1 PLd category 3 and EN ISO 10218-1 (UR collective datasheet). Street pricing is reported at roughly $45,000 to $60,000 for a UR10e and over $85,000 for a UR20, which is a secondary estimate (Standard Bots). Research-grade and low-cost alternatives exist with far weaker specification discipline: AgileX Piper is listed around 1.5 kg payload, 0.1 mm repeatability, 626 mm reach, 4.2 kg, at reseller prices from roughly $3,500 to $4,250, and Unitree Z1 pricing ranges across sources from about $11,800 to $20,799 with payload quoted inconsistently between 2 kg and 5 kg. Those inconsistencies are themselves the finding: the low-cost arm segment does not yet publish specifications you can design against.

Mobile bases and integrated mobile manipulators. MiR is the reference for bases, with the MiR250 and MiR600 figures given in section 6 and the MC600 pairing a MiR600 with a UR20 or UR30. Hello Robot's Stretch is the research-platform option, with Stretch 3 reported at $24,950 and about 54 pounds, and Stretch 4 introduced in May 2026 with a published 0.55 m reach, 4 kg maximum payload retracted and 2.5 kg extended, 46 kg total, shipping typically 2 to 4 weeks from purchase order, and explicitly certified only for laboratory and research use (Hello Robot Stretch 4 hardware guide; The Robot Report on Stretch 3). That research-only certification is a hard blocker for a commercial deployment and is the kind of line that gets missed. Boston Dynamics Stretch and Agility Digit are covered in section 2; neither publishes a list price, and reported Stretch figures of $300,000 to $500,000 are explicitly not a current manufacturer price.

Grippers and tooling. Robotiq (2F-85, 2F-140, Hand-E), OnRobot (2FG7, 3FG15, VGC10), Schmalz and Piab for vacuum, and Soft Robotics for food-adjacent soft grippers are the main commercial options, with specifications as given in section 3. Reseller prices seen during research were roughly $5,825 to $5,990 for a 2F-85 and roughly $3,782 to $6,030 across the OnRobot line, all third-party and not official list.

Dexterous hands. This is where the price data is weakest and where I would not quote a figure to a founder without a quote in hand. Shadow's Dexterous Hand is described by the manufacturer as 24 degrees of freedom with 20 motors (Shadow Robot), with secondary sources estimating anywhere from roughly $80,000 to $130,000 depending on configuration. The Wonik Allegro Hand V5 Sense is described as 16 degrees of freedom with pressure sensing, estimated around $35,000. Open-source and low-cost options include the LEAP Hand, assemblable from off-the-shelf parts for under $2,000, with Orca Hand around $6,000 and Inspire RH56 around $8,000, all secondary estimates. The practical read: a research-grade five-finger hand costs between ten and one hundred times a parallel-jaw gripper and brings tendon and calibration maintenance with it.

Tactile and force/torque. GelSight DIGIT is commercially available with a 4 to 6 week estimated lead time and replaceable gel cartridges, and GelSight states it is not intended for quantitative measurement (GelSight). GelSight Mini is the robot-mountable product, but its public page lists no numeric resolution, frame rate, dimensions, interface, or price, which is itself worth noting when planning a BOM (GelSight Mini). ATI, now part of Novanta, is the reference for wrist force/torque with the Axia80 at 16-bit resolution, Ethernet or EtherCAT, up to 8 kHz, IP60 (ATI Axia80). Robotiq's FT 300-S is the cobot-integrated alternative, IP65, with force range and noise figures that I could only find on a single reseller page and therefore do not consider verified.

Vision. Intel RealSense for low-cost depth (D405 for wrist, D435 and D455 for scene), Zivid and Photoneo (Zebra) for industrial structured light, with the specification-comparability caveat from section 7.

Systems integrators and picking-system vendors. RightHand Robotics, Osaro, Ambi Robotics, Dexterity, and Plus One Robotics are the incumbents in piece picking. Of these, Osaro markets explicitly to high-mix, high-variability fulfilment and bagging workflows, which is the nearest adjacent positioning to this deployment, though that is vendor marketing rather than a deployment report. During research I could not find any confirmed deployment of apparel polybag piece picking by any of these vendors, which is a meaningful negative result: if a major integrator had solved this profitably at scale, it would be in a press release.

Questions for Aqilis: Which components are they buying versus building, and why for each? What quotes do they actually hold, with what lead times? Which single component has the longest lead time, and what is the plan if it slips?

13. System-level economics

What is the correct unit of account?

Cost per successfully handled item, inclusive of exception handling, with damage treated as a separate and dominant term. Stated as a model:

$$\mathrm{CPSI} = \frac{\text{amortised capex} + \text{annual opex} + \text{exception labour} + \text{damage cost}}{\text{attempts/yr} \times \text{attempt rate} \times \text{success rate} \times \text{availability}}$$

Work the arithmetic for a plausible single intake cell. Take a $150,000 installed cell, which is roughly a UR10e plus tooling plus vision plus frame plus integration at the pessimistic end of the street estimates in section 12, amortised over three years at $50,000 a year. Run it 16 hours a day, 250 days a year, at a sustained 300 picks per hour, which is the Dex-Net 4.0 published rate and a defensible planning figure for suction-first picking on novel rigid items. At 95% success and 90% availability that yields 300 x 16 x 250 x 0.95 x 0.90, or about 1,026,000 successful items a year, and the capex term is therefore about $0.049 per item. Add maintenance, consumables, power, and a share of a technician, and the hardware cost per item is plausibly in the range of $0.10 to $0.20.

Now the term that dominates everything. The RealReal's average order value is $594 (SEC 2025 10-K). If the cell damages or downgrades 0.1% of items it handles, and the average value destruction on a damaged luxury item is conservatively a quarter of its value, the expected damage cost is 0.001 x 0.25 x $594, or about $0.15 per item, which is of the same order as the entire hardware cost. At 0.5% damage it is $0.74 per item and the cell is uneconomic regardless of how fast it runs. This is the number to put in front of Aqilis, because it inverts the usual optimisation target: for this customer, a system that runs at 200 picks per hour with a 0.05% damage rate beats one that runs at 400 with 0.3%, and no amount of throughput engineering recovers the difference.

Why is the cheapest robot rarely the cheapest deployed system?

Because the arm is a minority of the installed cost and essentially none of the risk. The costs that actually scale are integration engineering, custom tooling and fixturing, driver and software maintenance for every non-standard component, safety validation and documentation, commissioning, operator training, spares inventory, and the long tail of exception handling that persists after go-live. A $3,500 arm with no published repeatability under load, no safety certification, and no maintained ROS 2 driver does not save $40,000 against a UR10e; it moves that $40,000 into engineering time and schedule risk, and it adds a certification problem that may be unsolvable for a commercial deployment. The Hello Robot research-only certification noted in section 12 is the clean example of this category of hidden cost.

The second structural point is that exception handling does not go to zero, and its cost is set by how the exception is handled rather than how often it occurs. If a failed pick requires a human to walk to the cell, the marginal cost of an exception is several minutes of labour. If it requires a human to confirm an item's condition after a drop, it is several minutes of an authenticator's time, and authenticators are the expensive labour in this business. Designing the exception path is a higher-leverage activity than improving the success rate by a point.

What does the vendor's own economics argument look like?

Aqilis's white paper builds a three-year cost model with one-time pre-deployment integration cost plus four ongoing centres: acquisition, processing, storage, and residual review, with reliability following a saturating curve and residual review decaying toward zero as site adaptation absorbs corrections. Two features of that model are worth engaging with rather than accepting. It concedes, deliberately, that its processing and storage costs exceed teleoperation's, because rejecting pooled cloud retraining to preserve data sovereignty means running consolidation on constrained on-site hardware. That is an honest and unusual concession, and it has a direct hardware consequence: the on-site compute is a real capital line, not a rounding error, and someone should size it. And every figure in the model comes from their own internal projection rather than measurement, which the paper states. The marginal-cost-of-the-next-reliability-point framing is the right framing; the curve parameters are assumptions.

Questions for Aqilis: What is their current damage or downgrade rate on real items, however small the sample? What is their modelled cost per successfully handled item, and what attempt rate and availability does it assume? What on-site compute does the consolidation cycle require, and is that in the capex model?

14. Practical architecture recommendations

All three configurations below assume the same thing, which should be validated first: that the target station is item intake or induction at an authentication centre, that items arrive singulated or singulatable, and that a human expert remains in the loop for authentication and condition grading. Component choices name specific products to make the trade-offs concrete, not because those are the only valid choices.

(a) The simplest reliable MVP.

A fixed-base cell at one induction station. A UR10e for reach and payload headroom, bolted to a welded frame with the work surface fixtured at a known height. A combination end effector carrying both a suction channel and a parallel-jaw channel on one tool, avoiding tool-change time entirely: this is the RightHand Robotics architecture and it is the right shape for randomised single-item flow. An industrial structured-light 3D camera overhead for bin and tote perception, accepting the 300 to 800 ms acquisition cost, plus a RealSense D405 on the wrist for close-range re-localisation. An ATI Axia80 at the wrist, because the UR10e's built-in 5.5 N force accuracy cannot support a careful-handling claim. Fixed, controlled lighting. Scope restricted hard: polybagged and boxed items only, into totes, with everything else routed to the human path.

The trade-off is that this configuration does nothing novel and solves perhaps 40 to 60% of inbound by count, a figure that needs measurement rather than assumption. The bottleneck is perception on specular polybag film and transparent packaging, not manipulation. Before buying anything, run the item trial described below.

(b) The flexible system for varied inventory.

Add a second arm and make the cell bimanual, which is the single highest-value architectural change available because most of the hard tasks in section 1 are bimanual for humans too: holding a bag open while extracting, tensioning a garment between two grasps, stabilising an item while presenting a serial tag to a camera. Two UR arms on a common frame with a measured arm-to-arm calibration, each carrying a different end effector, so the system gets tool diversity without tool-change latency and without a changer's mechanical failure mode. Add tactile fingertips on the jaw channel only if slip detection is wired into a reflex rather than a log. Add a tool changer only if items can be batched into category runs, in which case it becomes clearly worthwhile.

Mobility belongs in this tier only if the task is storage retrieval rather than induction, in which case a MiR600 with a UR20 is the productised path, with the dock-error budget from section 2 closed by wrist perception. The trade-off is roughly double the capex and materially more calibration and safety-validation work, and the bottleneck shifts from perception to bimanual coordination and to the integrated safety case.

(c) The advanced dexterous system.

Dual arms with at least one multi-finger hand, tactile sensing across the fingertips, on-site compute sized for the consolidation cycle, and a demonstration-collection path. This is the configuration that could plausibly attack unbagging, hanger insertion, and laying a garment flat, which are the tasks that would actually displace significant labour at The RealReal. It is also, on the evidence in sections 3, 4, and 10, a research programme rather than a deployment: the cloth manipulation literature is still working on towels, vision-based tactile sensing is below the bandwidth that slip arrest requires, and simulation cannot yet predict deformable contact outcomes. If Aqilis is building this, the honest framing to them is that the hand is a bet on a capability that does not exist yet in anyone's shipped product, and that the deployment should not depend on it.

What should be tested before any hardware is purchased?

One test, and it is cheap, fast, and almost nobody runs it. Take a stratified sample of 100 real items from actual inbound, proportional to real category mix: handbags, shoes boxed and loose, garments bagged and loose, small leather goods, jewellery, watches, and whatever the long tail contains. For each candidate end effector, attempt the grasp by hand with instrumentation: a vacuum gauge on the suction line and a force gauge or scale on the lift. Record for every item whether a seal was achieved, the hold force, the hold duration, whether the grasp marked the item, and whether it was ungraspable at all. The output is a per-category attempt rate and a per-category failure mode list, obtained in a few days for the cost of a gauge, and it determines the end effector choice, the achievable attempt rate, and therefore the entire economic model. Any projection built before that test is a guess.

The secondary pre-purchase tests, in priority order: measure the actual dock-to-grasp pose error if mobility is proposed; measure the structured-light scan time on real specular packaging rather than on a calibration target; measure the teleoperation round trip on the real network path if a collection rig is planned; and obtain written confirmation of safety certification status for every component intended for a commercial, non-research deployment.

A. One-page hardware decision cheat sheet

Start from the task, not the platform. Decompose to primitives, then ask of each: does it need two hands, does it need in-hand reorientation, does it need to be mobile. Most answers are no, no, no.

Default to fixed-base. Add mobility only when the work cannot be brought to the robot. Mobility costs about 3 mm and 0.5 degrees of pose error, a second safety regime, and battery logistics.

Default to suction first, jaw second, hand last. Suction was chosen for 82% of grasps when a policy was free to choose. A five-finger hand is 10 to 100 times the cost of a gripper and is justified only by genuine in-hand manipulation.

Combination tool for randomised flow, tool changer for batched flow. The change cost amortises within a category run and never amortises in a randomised one.

Read payload as payload at full CoG offset across the whole workspace, minus the tool mass, minus the camera. Read reach as reach to the flange, minus tool length.

Built-in flange force sensing is not careful-handling sensing. A UR10e resolves 5.5 N. If the value proposition is not marking a handbag, buy a real wrist sensor.

Slip detection needs bandwidth, not resolution. A 60 Hz vision-based tactile sensor observes slip; it does not arrest it.

Depth sensor selection is a working-distance decision. 7 to 50 cm for wrist, 0.6 to 6 m for scene, structured light for the bin at 300 to 800 ms per scan. Never put a 0.4 m minimum-range camera on a wrist.

LiDAR is navigation hardware. If it appears in a manipulation BOM, ask which decision it informs.

Controlled lighting is the cheapest perception upgrade available and is routinely the highest leverage.

EtherCAT for synchronised multi-axis, CAN for a gripper, Ethernet for the cell. Sub-100 microsecond jitter is achievable with PREEMPT_RT, CPU isolation, and a dedicated bus thread, and not otherwise.

Safety certification does not compose. PLd plus PLd does not equal PLd. Name the owner of the integrated safety case in writing, early.

Research-only certification is a commercial blocker, not a footnote.

Measure attempt rate and success rate on attempts separately. A 98% success rate on the easy 60% is a 59% system.

For this customer, damage rate dominates throughput. At $594 average order value, 0.1% damage costs about the same per item as the entire amortised hardware. Optimise the wrong variable and no throughput gain recovers it.

Run the 100-item trial before buying anything. A vacuum gauge, a force gauge, and three days beats any simulation.

B. Comparison tables

End effector families. Costs are street estimates unless noted; confirm by quote.

Family	Representative products	Typical grip range / force	Item classes it wins on	Fails on	Street cost	Reliability notes
Suction, single or multi-cup	Schmalz Bin-Picker SBPG, Piab cups, custom arrays	Lift force set by cup area x pressure differential; Schmalz bag tooling to 50 kg over 405 x 305 mm	Polybags, boxes, flat sealed surfaces, smooth leather panels	Porous knits, mesh, perforated or heavily textured surfaces, curved small items	Low, hundreds to low thousands	Cups are consumables; dust degrades seal; needs generator and valve sizing
Parallel jaw	Robotiq 2F-85, OnRobot 2FG7	2F-85: 85 mm stroke, 20 to 235 N; 2FG7: 38 mm stroke, 20 to 140 N, 11 kg form fit, IP67	Fabric, straps, small rigid items, anything suction cannot seal	Large flat items, items needing area support, fragile items if minimum force is coarse	~$4,800 to $6,000	Mature, sealed options available, few failure modes
Three-finger centric	OnRobot 3FG15	150 mm stroke, 4 to 152 mm external, 15 kg form fit	Cylindrical and rotationally symmetric: bottles, watch cases, rolled goods	Flat or irregular items	~$6,000	Mature
Adaptive / underactuated	Robotiq adaptive line, soft grippers	Conforms to shape, force-limited	Unknown geometry, mixed clutter	Tasks needing a specific contact configuration	Low to mid thousands	Good shape tolerance, poor controllability
Combination suction + jaw	RightHand Robotics RightPick	Vendor best case up to 1,200 items/hour, stated as a best case	Randomised single-item flow across mixed classes	Tasks needing two hands or in-hand reorientation	System-level, quote only	The shipped architecture for piece picking
Multi-finger dexterous	Shadow Dexterous Hand (24 DoF, 20 motors,	Fingertip-precision and in-hand reorientation	Unbagging, clasp manipulation,	Everything simpler, on cost and	LEAP under $2,000; Orca	Tendon wear, calibration burden,

Family	Representative products	Typical grip range / force	Item classes it wins on	Fails on	Street cost	Reliability notes
	manufacturer), Wonik Allegro V5 Sense (16 DoF), LEAP Hand		in-hand reorientation	reliability grounds	~$6,000; Inspire ~$8,000; Allegro ~$35,000; Shadow ~$80,000 to $130,000, all estimates	immature in deployment

Mobile manipulator configurations.

Configuration	Representative build	Pose error at tool	Payload budget	Cycle-time character	Safety maturity	Best fit here
Fixed-base single arm	UR10e on welded frame	~1 to 2 mm, calibration-dominated	12.5 kg minus tool	Fastest; no dock, no re-localisation	Highest; arm certified PLd Cat 3, EN ISO 10218-1	Induction and intake stations: the recommended MVP
Fixed-base dual arm	Two UR arms, common frame	~1 to 2 mm per arm plus arm-to-arm residual	2 x 12.5 kg minus tools	Slightly slower per arm, far more capable per cycle	High, but integrated safety case is bespoke	Bimanual intake: hold-and-extract, tension-and-flatten
Wheeled mobile manipulator	MiR600 + UR20 (MC600); MiR250 + UR5e	~8 to 12 mm, dock-dominated, needs wrist re-localisation	Base 600 kg transport; arm 20 kg minus tool	Adds dock plus re-localisation per location	Good; ISO 3691-4, R15.08-1 on base, system case bespoke	Storage retrieval across racking, not induction
Purpose-built logistics MM	Boston Dynamics Stretch	Not published; vendor-managed	23 kg max case	600 to 800 cases/hour, vendor	Shipped product, CE declared	Container and trailer unload only; object class is cardboard boxes 6 to 36 inches
Research mobile manipulator	Hello Robot Stretch 3 / 4	Not published	Stretch 4: 4 kg retracted, 2.5 kg extended	Low speed, low payload	Stretch 4 certified for laboratory and research use only	Prototyping and data collection; not a commercial deployment
Humanoid	Agility Digit (35 lb, 4 h); Digit 5 announced 50 lb, 20 h, early access H1 2027	Not published	16 to 23 kg	Vendor-managed; GXO cited ~98% accuracy on task	Current units in segregated zones; Digit 5 marketed as cooperatively safe	Not recommended for fragile luxury goods near experts in 2026

C. Ten high-value questions for Aqilis

1. Which station in The RealReal’s intake flow are you automating, who chose it, and what fraction of the site’s labour hours does it represent?

2. On a stratified sample of real inbound items, what is your attempt rate, and what is your success rate on attempts? If you have not run that sample, what is blocking it?
3. What is your damage or downgrade rate on real items, and how is damage detected rather than assumed absent?
4. What force resolution do you have at the contact, and is it from the arm's flange or a dedicated wrist sensor? If flange-only, how do you support a careful-handling claim at 5 N granularity?
5. Your white paper specifies a sub-20-millisecond reflexive safety shield regulated by control barrier functions. What hardware executes it, on what kernel and thread, measured how, and is it safety-rated or advisory?
6. The architecture requires a hardware-level isolation switch severing actuator control paths for the sleep cycle. Is that a safety-rated contactor, and who validated it?
7. The sleep cycle presupposes scheduled facility downtime. Reverse logistics at peak season may have none. What happens to the consolidation loop in November and December, and does the capex model include the on-site compute it needs?
8. Section 3.4 claims third-person video replaces teleoperation. What is the current empirical evidence, and what is the fallback data-collection rig and its cost if the result does not hold?
9. Who owns the integrated safety case for the deployed cell, and has a notified body or safety consultancy been engaged? Which components are certified for commercial rather than research use?
10. k_consensus and tau_novelty are the free parameters on which the promotion and capture story rests. What values are you using, how were they chosen, and what happens at the wrong value?

D. Prioritized study plan

Before the meeting, in priority order:

- ☐ Re-read your own brain page on the Aqilis CAE white paper, specifically sections 3.4, 3.5, 4.3, and the appendix latency tiers, so you can ask question 5 and question 8 precisely.
- ☐ Read the Boston Dynamics Stretch specification sheet end to end as an exercise in how a shipped mobile manipulator scopes its object class (brochure).
- ☐ Read the Universal Robots collective datasheet force-sensing rows and commit the UR10e numbers to memory: 100 N range, 5.5 N accuracy (datasheet).
- ☐ Skim Dex-Net 4.0 for the headline result and the 82% suction figure (Science Robotics).
- ☐ Read The RealReal's 2025 10-K Item 1 on the authentication workflow and the key metrics table (SEC).
- ☐ Work the cost-per-item arithmetic in section 13 yourself with your own assumptions, so you can defend or revise the damage-dominance conclusion live.

Over the following two weeks:

- ☐ Read the Frontiers 2026 systematic review of deep-learning cloth manipulation, including its comparison tables (Frontiers).
- ☐ Read the CLASH paper and the Hard Contacts with Soft Gradients paper for the contact-model limits (CLASH; arXiv 2506.14186).

- ☐ Obtain and read ISO 10218-2:2025, specifically the absorbed body-region force and pressure tables, rather than relying on vendor summaries.
- ☐ Obtain ANSI/A3 R15.08 Parts 2 and 3 and read the integrator and user obligation clauses.
- ☐ Build a reach and collision model of one candidate cell in MuJoCo or Isaac and measure how much the structured-light scan time dominates the cycle.
- ☐ Price a full BOM for configuration (a) by actually requesting quotes, so you learn the real lead times rather than the published ones.
- ☐ Run or specify the 100-item grasp trial from section 14, which is the highest-information experiment available in this engagement.

E. Glossary

AMR / IMR. Autonomous mobile robot; industrial mobile robot, the term used by the ANSI/A3 R15.08 standard series.

Attempt rate. The fraction of presented items a system tries to handle, as distinct from its success rate on those attempts. Reporting only the latter inflates apparent performance.

Backdrivability. Whether an external force at the output can drive the joint backwards through its transmission. High-ratio geared joints are not backdrivable; quasi-direct drive is.

Backlash. Lost motion on reversal in a transmission, measured in arcminutes or arcseconds. A day-one specification that degrades with wear.

CoG offset. Centre-of-gravity offset of the payload from the tool flange. Payload ratings are only meaningful when stated at full CoG offset across the workspace.

Control barrier function (CBF). A constraint-based method for guaranteeing a system stays in a safe set, used in Aqilis’s architecture as the analytical safety shield.

Cycloidal / RV reducer. A reduction mechanism distributing load across many contact points, tolerant of shock, with somewhat higher new-condition backlash than strain-wave gearing.

Force closure. A grasp whose contacts can jointly resist any external wrench, relying on friction. Form closure is the stronger, geometry-only version.

Friction cone. The set of contact forces that do not cause slip, a cone of half-angle arctan(mu) about the surface normal.

Hand-eye calibration. Solving for the rigid transform between a camera and a robot frame, classically formulated as AX = XB.

Harmonic / strain-wave drive. Compact high-ratio reduction using a flexible spline, giving very low backlash at the cost of shock tolerance.

MPPH. Mean picks per hour, the standard throughput metric for picking systems. Always ask what object set and what success definition produced it.

MTBF / MTTR. Mean time between failures and mean time to repair. Availability equals MTBF divided by (MTBF + MTTR), and MTTR is usually the mismanaged term.

PLd Category 3. A performance level and architecture category from ISO 13849-1 describing required safety-function reliability and fault tolerance. Does not compose across components.

Power and force limiting (PFL). The collaborative mode in which contact with a human is permitted but bounded by body-region force and pressure limits. A validated claim, not a design intent.

Quasi-direct drive (QDD). Low-ratio, high-torque-density actuation prioritising backdrivability and bandwidth over precision and holding torque.

Quasi-static vs transient contact. Clamping against a fixed surface versus brief impact. Quasi-static limits are lower, and a tool can pass one test and fail the other.

Series-elastic actuator (SEA). An actuator with a deliberate compliant element whose deflection is measured to obtain joint torque, trading bandwidth for shock tolerance and true torque control.

Singulation. Separating a stream or pile of items into individually handleable units.

Wrench. A combined force and torque vector, the natural six-dimensional unit for grasp analysis.

Closing: what you now know, what to study, what I recommend

What you now know. The RealReal's manipulation problem is structurally unlike the fulfilment problems that shaped the piece-picking industry, because its inventory is unique per item, high in value, and graded on condition, so the economics are governed by damage rate rather than throughput: at an average order value of $594, a 0.1% damage rate costs roughly as much per item as the entire amortised hardware, and at 0.5% nothing recovers it. The task decomposes into a feasible tier (polybagged and boxed items into totes, transport, presentation to a scanner) worth perhaps 40 to 60% of inbound by count, a marginal tier (garments from piles, bags with hardware, loose shoes), and a hard tier (unbagging, hanger insertion, laying flat, jewellery) where the academic literature is still working on towels and no integrator has a confirmed deployment. Architecturally, mobility costs about 3 mm and 0.5 degrees of pose error that only wrist perception recovers, which argues for fixed-base cells wherever conveyance already brings work to a station, and bimanual capability is the highest-value upgrade because the hard tasks are bimanual for humans too. On end effectors, suction carries the majority of the work and a free-choosing policy picked it 82% of the time, a parallel jaw covers what suction cannot seal, and a five-finger hand costs ten to one hundred times more and is justified only by genuine in-hand manipulation. The sensing conclusion is sharp: a UR10e's built-in flange force sensing resolves 5.5 N, which cannot support a careful-handling claim, so a dedicated wrist force/torque sensor is a requirement rather than an upgrade, and vision-based tactile sensors at tens of hertz observe slip rather than arrest it. Integration fails not at flanges and connectors but at synchronised timing, recurring calibration, and above all at the fact that safety certification does not compose, so a PLd arm on a PLd base is not a PLd system until somebody designs, validates, and signs

the integrated case. Simulation is reliable for geometry, reach, and cycle time and unreliable for exactly the contact physics that decides whether a handbag rotates out of a grasp, with torsional friction specifically identified as a current failure of MuJoCo's contact model. And the market's pricing information is poor enough that most figures in circulation are secondary estimates from aggregators, which makes "what quotes do you actually hold" a more informative question than any spec comparison.

Checklist, what is worth studying next. This is in Appendix D, split into the six items worth doing before the meeting and the seven worth doing in the following two weeks. If you do only one thing from it, read the UR force-sensing rows and the Stretch object-class scoping, because together they let you ask the two questions that most quickly reveal whether a robotics founder has engaged with physical reality.

My recommendations. First, go into the meeting arguing for scope reduction rather than capability expansion, and bring the Stretch specification sheet as the precedent: Boston Dynamics shipped a mobile manipulator by constraining the object class to taped cardboard boxes between 6 and 36 inches, and that discipline is why it ships. Aqilis's beachhead reasoning is sound, because reverse logistics genuinely does supply the high-salience edge cases their architecture is built to consume, but the same messiness that makes it a good data environment makes it a poor first deployment target, and those two facts need separating explicitly. Second, push hard for the 100-item grasp trial before any hardware purchase, because it costs a vacuum gauge and three days and it determines the attempt rate, the end effector, and the entire economic model; any BOM specified before that test is a guess with a price attached. Third, advise configuration (a) as the deployment and configuration (c) as the research track, funded and measured separately, because conflating them is how a startup ends up with a dexterous hand on a customer site and no reliable pick. Fourth, raise the damage-dominance arithmetic early and in the customer's own units, since it reframes the engineering target and is the single most valuable thing an outside advisor can contribute to this account. Fifth, on their architecture specifically, the two questions I would not let go unanswered are what hardware executes the sub-20-millisecond safety shield and whether it is safety-rated, and what happens to the sleep-cycle consolidation loop during peak season when the facility does not stop; the first is a certification question that gates commercial deployment, and the second is a hole in the architecture's core assumption that their own paper does not address. Finally, hold the line on the distinction between their white paper's internally modelled figures and measured results. The paper is explicit that every cost figure comes from their own model, and the right posture is to treat the framing as valuable and the curve parameters as assumptions, which is a more useful form of support than either credulity or dismissal.

Provenance note: this primer draws on the Aqilis Research white paper "Continual Adaptation at the Physical Edge" (dated 19 September 2026, patent pending, read via share.aqilis.ai and summarised in the private brain page media/papers/aqilis-continual-adaptation-engine) in addition to the public manufacturer specifications, standards documents, and peer-reviewed and preprint sources linked inline. Figures attributed to manufacturers are from their own published specifications; figures labelled as estimates come from resellers, aggregators, or secondary commentary and were not confirmed by quote.